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import java.util.Scanner;

abstract class Tour {
    protected int tourId;
    protected String tourName;
    protected int tourDuration;
    protected double tourCost;

    public Tour(int tourId, String tourName, int tourDuration, double tourCost) {
        this.tourId = tourId;
        this.tourName = tourName;
        this.tourDuration = tourDuration;
        this.tourCost = tourCost;
    }
}

abstract class AbstractTour extends Tour implements BookingSystem {
    protected String tourGuideName;
    protected int minimumGroupSize;

    public AbstractTour(int tourId, String tourName, int tourDuration, double tourCost,
String tourGuideName, int minimumGroupSize) {
        super(tourId, tourName, tourDuration, tourCost);
        this.tourGuideName = tourGuideName;
        this.minimumGroupSize = minimumGroupSize;
    }

    public abstract double calculateTotalCost();

    public void displayTourDetails() {
        System.out.println("Tour ID: " + tourId);
        System.out.println("Tour Name: " + tourName);
        System.out.println("Tour Duration: " + tourDuration + " days");
        System.out.println("Tour Cost: $" + tourCost);
        System.out.println("Tour Guide: " + tourGuideName);
        System.out.println("Minimum Group Size: " + minimumGroupSize + " people");
    }

    public void bookTour() {
        System.out.println("Booking tour: " + tourName);
        // Perform booking logic here
        System.out.println("Tour booked successfully!");
    }

    public void cancelTour() {
        System.out.println("Canceling tour: " + tourName);
        // Perform cancellation logic here
        System.out.println("Tour canceled successfully!");
    }
}

class AdventureTour extends AbstractTour {
    private String activityLevel;

    public AdventureTour(int tourId, String tourName, int tourDuration, double tourCost,
String tourGuideName, int minimumGroupSize, String activityLevel) {
        super(tourId, tourName, tourDuration, tourCost, tourGuideName, minimumGroupSize);
        this.activityLevel = activityLevel;
    }
}

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    public double calculateTotalCost() {
        // Add specific calculations for AdventureTour's total cost
        return tourCost;
    }
}

class CityTour extends AbstractTour {
    private String city;

    public CityTour(int tourId, String tourName, int tourDuration, double tourCost, String
tourGuideName, int minimumGroupSize, String city) {
        super(tourId, tourName, tourDuration, tourCost, tourGuideName, minimumGroupSize);
        this.city = city;
    }

    public double calculateTotalCost() {
        // Add specific calculations for CityTour's total cost
        return tourCost;
    }
}

class WildlifeTour extends AbstractTour {
    private String wildlifeType;

    public WildlifeTour(int tourId, String tourName, int tourDuration, double tourCost,
String tourGuideName, int minimumGroupSize, String wildlifeType) {
        super(tourId, tourName, tourDuration, tourCost, tourGuideName, minimumGroupSize);
        this.wildlifeType = wildlifeType;
    }

    public double calculateTotalCost() {
        // Add specific calculations for WildlifeTour's total cost
        return tourCost;
    }
}

interface BookingSystem {
    void bookTour();

    void cancelTour();
}

class TourApp {
    public static void main(String[] args) {
        try {
            AdventureTour adventureTour = new AdventureTour(1, "Adventure Tour", 7, 1000.0
, "John Doe", 5, "High");
            CityTour cityTour = new CityTour(2, "City Tour", 3, 500.0, "Jane Smith", 10, "
New York");
            WildlifeTour wildlifeTour = new WildlifeTour(3, "Wildlife Tour", 5, 1500.0, "
David Johnson", 8, "Safari");

            // Test methods of AdventureTour
            adventureTour.displayTourDetails();
            adventureTour.bookTour();
            adventureTour.calculateTotalCost();

            System.out.println();

            // Test methods of CityTour

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        cityTour.displayTourDetails();
        cityTour.bookTour();
        cityTour.calculateTotalCost();

        System.out.println();

        // Test methods of WildlifeTour
        wildlifeTour.displayTourDetails();
        wildlifeTour.bookTour();
        wildlifeTour.calculateTotalCost();

        System.out.println();
    } catch (Exception e) {
        System.out.println("An error occurred: " + e.getMessage());
    } finally {
        System.out.println("Tour app execution completed.");
    }
}
```